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Non-Adaptive Scheduling of Entanglement Purification for All-Photonic Quantum Repeaters

Author Eliott Flechtner^{*}

Supervisors Prof. Rodney Van Meter^{†§}

Proj. Assoc. Prof. Michal Hajdušek^{†§}

Proj. Assist. Prof. Naphan Benchasattabuse[†]

Laboratory AQUA – Advancing Quantum Architecture, Keio SFC

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^{*}*M2 QMI, Institut Polytechnique de Paris & EPITA, France*

[†]*Graduate School of Media and Governance, Keio University, Kanagawa, Japan*

[§]*Quantum Computing Center, Keio University, Kanagawa, Japan*

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Abstract

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Long-distance quantum networks require repeaters to distribute high-fidelity Bell pairs despite photon loss and operational noise. In half-repeater-graph-state (HRGS) all-photonic repeaters, entanglement purification can be applied during resource generation, after a single hop, or after an end-to-end pair is established. Selecting where to purify, how many independent copies to consume, and which circuit to apply is therefore a combinatorial scheduling problem.

This thesis formalises every fixed, non-adaptive purification schedule as a directed acyclic graph and evaluates it with one bottom-up pass that computes fidelity, success probability, latency, rate, and generation-resource cost. The evaluator reproduces the source protocol’s three canonical fidelity benchmarks to four significant figures. On top of this model, the work implements structured baseline enumeration, a span-partition dynamic program, same-span pumping, and beam search. All returned candidates are structurally validated schedules evaluated by the same physical model.

At the source protocol’s nominal configuration, the matched-cost searched schedule improves rate by 2.5% over the hand-designed baseline. Allowing the search to use less than the reference budget yields a valid schedule at 60% of that budget with a 52.8% higher rate. More importantly, when inner-qubit error is introduced, the fixed baseline falls below the required fidelity while a searched schedule restores feasibility within the same resource budget. Additional experiments characterize noise sensitivity, timing effects, and the minimum searched resource budget as the chain grows.

These results demonstrate the value of systematic schedule selection within the implemented model. They do not establish global optimality: default pumping and beam search retain bounded frontiers, and a schedule not found is only absent from the families explored by the corresponding search.

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Chapter 1

Introduction

1.1 Motivation

Realizing a Quantum Internet [1], [2], i.e. a network connecting quantum devices over long distances, would enable applications with no classical equivalent [3], [4]. **Quantum Key Distribution (QKD)** provides a shared secret key whose security is guaranteed unconditionally by the principles of quantum mechanics, independently of an eavesdropper’s computational capabilities [5]–[7]. **Distributed Quantum Computing (DQC)** lets several small quantum processors, connected by such a network, act together as one larger one [8], [9]. Finally, networked quantum sensing can improve measurement precision beyond what independent classically connected sensors can achieve [3], [10], [11]. These applications rely on distributed entanglement as a shared quantum resource; in the repeater architectures considered here, this resource is represented by high-fidelity Bell pairs, also called *ebits*. Direct photon transmission cannot supply this resource over long distances because channel loss grows exponentially with length [3]. Quantum repeaters address this limitation [12]; Chapter 2 reviews how successive repeater generations do so.

Regardless of repeater generation, the delivered entanglement is noisy. Entanglement purification is one standard way to turn such noisy entanglement into a usable resource: it consumes several independently generated copies of a shared pair and returns fewer copies of higher fidelity [13], [14]. Each round trades additional resource consumption for a probabilistic fidelity gain. In a multi-hop repeater chain, purification can be placed at several stages: while resources are generated, after individual links are established, or at the end nodes. The number of copies consumed and the purification circuit may also vary from one stage to another. This thesis studies an all-photonic repeater architecture built from **Repeater Graph States (RGSs)** [15], in which purification can be inserted at different stages of resource generation, link establishment, and end-to-end connection rather than only after end-to-end entanglement is established. Choosing how to allocate a fixed purification budget across this chain is therefore a combinatorial scheduling problem. The central question is which schedule gives the best specified trade-off between rate, fidelity, and resource cost.

1.2 The Scheduling Gap

Two recent papers by Benchasattabuse, Hajdušek, and Van Meter establish the specific architecture and scheduling flexibility this thesis builds on. Throughout this thesis, [16] is called *the Bridging paper* and [17] is called *the Integrating paper*. Together, they are called *the source papers*.

The Bridging paper introduces the **Half-RGS (HRGS)** building block: half of an **RGS**, generated and held at a single station rather than requiring the full state to exist in one place, allowing an end node to act as its own source of entanglement. This enables the all-photonics scheme to interoperate with conventional memory-equipped repeater nodes instead of requiring every intermediate station to be purely photonic [16]. The Integrating paper then shows that entanglement purification can be inserted into this architecture at any stage, generation, link, or end node, and mixed freely between stages, provided it is executed *optimistically* [17], [18], i.e. without waiting for a classical heralding outcome between rounds. Chapter 3 formalises why this timing model is what makes purification worth inserting at all.

The Integrating paper demonstrates this flexibility with a single chosen schedule and identifies the search for a better one, under a fixed physical network and resource budget as an open problem. It does not establish which allocation of a fixed budget performs best, or whether that answer changes with the noise model or chain length. This is the scheduling gap this thesis addresses. Given a fixed network configuration, the research problem is to select from the combinatorially many schedules that differ in purification location, copy count, circuit, and order, then optimise a stated objective such as rate, fidelity, or cost. The working hypothesis is that systematic search finds schedules that improve on the hand-picked baseline at equal cost and, more importantly, retain feasibility in regimes where that baseline fails. This thesis tests that hypothesis across several network configurations beyond the source papers' reference configuration.

1.3 Research Objectives and Contributions

Addressing this gap requires a precise object to search over, since neither source paper formalises the schedule space, as well as search algorithms capable of exploring that exponentially large space. This thesis makes four concrete contributions toward that end.

1. A formal, **Directed Acyclic Graph (DAG)**-based model of the schedule space for **HRGS**-based quantum repeaters (Chapter 4) and a bottom-up evaluator that reproduces the source paper's fidelity benchmarks for the raw, heralded, and optimistic canonical schedules to four significant figures (Chapter 5).
2. A three-tier search framework: structured brute-force baselines, an exact span-partition dynamic program, and bounded same-span pumping and beam-search extensions (Chapter 5). Every returned schedule is structurally validated and evaluated using the same physical model. At the paper's reference configuration, the matched-cost result improves rate by 2.5%, while a budget-relaxed result reaches a rate of 6195.95 (the throughput metric $R(\Sigma)$)

defines in Chapter 4) at cost 60, a 52.8% improvement while only using 60% of the paper baseline’s budget (Chapter 6).

[recheck after said chapters are written, results may change:

3. An empirical study of when scheduling changes feasibility rather than only quality, varying outer- and inner-qubit noise, the number of Bell-State Measurement (BSM) arms per hop, hop count, memory decoherence, and generation timing (Chapters 6–7). Its strongest result is a fixed-budget feasibility advantage: under non-ideal inner-qubit error, the paper schedule falls below the $F \geq 0.9$ threshold while a searched schedule at the same budget still clears it.]

[same as above:

4. Five named design findings extracted from that characterization (Chapter 7). In particular, within the searched families the minimum feasible budget over $N = 10\text{--}28$ is described by $e_{\text{max}}^{\text{min}} \approx 1.369 N^{1.620}$. This is a descriptive fit, not an asymptotic law, and its crossover with the paper’s linear budget formula $10N$ is beam-width-sensitive.]

Every result above is an existence claim within the schedule families actually searched, not a proof of global optimality; Section 7.4 returns to this distinction.

1.4 Thesis Outline

The remaining chapters proceed as follows. Chapter 2 surveys quantum repeater protocols, entanglement purification, and RGSs architectures, positioning this thesis relative to that state of the art. Chapter 3 introduces the physical objects which the rest of the thesis builds on: Bell pairs and the HRGS architecture, the error-vector and side-effect representations used to track a resource’s quality through a schedule, and the purification circuits and optimistic execution model from the Bridging and Integrating papers. Chapter 4 turns these objects into a formal and searchable model: the network configuration, the state carried by each partially built resource, the schedule itself as a DAG of operations, the resource budget, and the optimization objectives this thesis considers. Chapter 5 describes the shared evaluator and the three search tiers built on top of that model. Chapter 6 reports the experimental setup, validates the evaluator against the source papers, and presents the search results across the tested regimes. [to review after chapter is written: Chapter 7 draws out the five design findings, discusses when scheduling matters and when it does not, and states the scope within which the reported results should be read as optimal and the limitations of this work.] Chapter 8 summarises the contributions and outlines directions for future work.

Chapter 2

Related Work and State of the Art

This chapter surveys the three bodies of work this thesis draws on: quantum repeater protocols in general, entanglement purification specifically, and **RGS**-based architectures, before positioning the scheduling gap identified in Section 1.2 against that state of the art.

2.1 Quantum Repeater Protocols

Sending a photonic qubit directly through an optical fiber or free space is limited by the channel's transmissivity, i.e. the probability that a photon successfully reaches the receiver. For a channel of length L and attenuation coefficient α , this is given by $\eta = e^{-\alpha L}$. At the typical telecom C-band attenuation of $\alpha_{\text{dB}} \approx 0.2$ dB/km [19], the transmissivity is already as low as $\eta \sim 10^{-40}$ over 2000 km. Thus, even with perfect single-photon sources and detectors, direct transmission becomes impractical over long distances, with the achievable rate decreasing exponentially with distance [20]. More generally, the *repeaterless secret-key capacity* of a lossy channel, bounded by the **Pirandola-Laurenza-Ottaviani-Banchi Bound (PLOB)**, exhibits the same exponential scaling in the high-loss regime [3], [21], [22].

A quantum repeater [12] divides a long channel into shorter elementary links, for which entanglement establishment is more feasible, then stitches the resulting short-range entanglement into a single end-to-end pair by entanglement swapping [23], [24]. Repeater proposals are commonly grouped into three generations according to how they address photon loss and operational errors along the chain [3], [25]. First-generation (1G) repeaters address both through **Heralded Entanglement Generation (HEG)** and entanglement purification [13], [14] (Section 2.2), storing successfully generated entangled pairs in matter-qubit memories while other links are established [12], [26]. Second-generation (2G) repeaters replace entanglement purification with **Quantum Error Correction (QEC)** to protect against operational errors, while still relying on heralding to detect photon loss [27], [28]. Third-generation (3G) repeaters use quantum error-correcting codes [29], [30] to protect against both loss and operational errors, avoiding the need to herald successful transmission across each elementary link [15], [31].

The all-photonic quantum repeaters [15] studied in this thesis belong to the third generation but use a different approach from most proposals in that category, which rely on matter-qubit error correction. Rather than protecting a stationary matter qubit, they encode the transmitted qubit itself in a multi-photon graph state [32]–[34]: the **RGS** [15]. The tree encoding lets the

protocol tolerate loss while operating on flying qubits, so no quantum memory is required at intermediate stations. Section 2.3 places this family among related architectures; Chapter 3 gives the mechanics needed for this thesis.

A related but architecturally distinct proposal, the measurement-based quantum repeater [35], keeps small pre-prepared entangled resource states [36]–[40] at each station and couples them to incoming photons with simple BSMs, rather than building one graph state that spans the whole repeater chain [41]–[43]. This design’s emphasis on minimising the qubit count held at each station is a useful point of comparison for the resource costs studied in Chapter 6.

2.2 Entanglement Purification Protocols

Chapter 1 already introduced the basic idea of trading copies for fidelity; this section covers the protocols that implement it. The two foundational two-qubit protocols, both using only Local Operations and Classical Communication (LOCC), are BBPSSW [13], which operates on Werner states, and DEJMPS [14], which extends the same approach to general Bell-diagonal states without first depolarizing them into Werner states through random local Pauli operations. Both protocols consume two copies and return one, succeeding only under the appropriate local measurement outcome condition: two units of resource cost are consumed to produce one output pair with increased fidelity.

Entanglement pumping describes a related idea: rather than purifying once and stopping [12], [44], the same working pair is purified repeatedly, or *pumped*, against a sequence of freshly generated pairs, increasing its fidelity further with each round [13], [14], [44]. Both ideas, the resource/fidelity trade-off and pumping, recur in the RGS-specific circuits on which this thesis’ formal model is built (Section 3.3.1).

Later work treats the purification circuit itself as an object to optimise rather than fixing it in advance. These approaches search directly over local Clifford operations to find shorter circuits with higher success probability and better output fidelity than the BBPSSW/DEJMPS family [45], particularly when finite resources and imperfect local operations are taken into account. Purification has also been generalised from bipartite Bell pairs to multipartite graph states [46], [47]. The RGS-specific two-to-one circuits used by the Integrating paper (Section 3.3.1) adapt the same purification principles to the RGS architecture rather than directly implementing the original BBPSSW or DEJMPS protocols.

Photon loss tolerance Purification addresses noise, not photon loss. Loss tolerance is nevertheless frequently discussed alongside purification in the photonic-repeater literature, so the distinction matters here. Tree encoding [48] provides a different form of redundancy: rather than consuming multiple copies of the same entangled pair to improve fidelity, it encodes one logical qubit in a tree-shaped graph state spanning multiple physical photons, whose branching factors set the achievable loss tolerance. The logical measurement outcome can then be reconstructed from the surviving branches. This approach was originally introduced for one-way measurement-based quantum computation [36], [38], [48]. It was later specialized into a loss-tolerant quantum memory scheme [49], along with an analysis of the resource-efficiency threshold at which such

encoding becomes practical [50]. It is this tree encoding, rather than purification, that allows the inner qubits of the RGS to tolerate photon loss (Section 3.1.2).

2.3 Repeater Graph State Architectures

The RGS was proposed as the resource state for the first all-photonic quantum repeater: a graph state whose outer qubits provide the entangling links to neighbouring stations and whose inner qubits are tree-encoded [48], allowing the logical connection to survive despite photon loss [15]. Later work has examined two recurring questions: how efficiently the RGS itself can be generated, and how much photon loss and operational error it can tolerate for a given resource budget. For generation, the original proposal constructs the graph state by fusing small entangled photon sets using linear optics [15], [22], while deterministic emitter-based schemes instead build the graph state directly from a small number of photon emitters which reduces the physical device count required per station [51]–[53]. These approaches have since been demonstrated experimentally at small scale [54]. For loss and error tolerance, alternative graph topologies and adaptive measurement strategies can improve the entanglement rate for a fixed loss and error budget [55]–[57]; code-concatenated variants instead address loss and operational errors using two specialized codes rather than a single tree code [28].

The Bridging and Integrating papers, discussed in Section 1.2 and formalised in Chapter 3, provide the specific starting point for this thesis: the former introduces the HRGS building block [16], [58], [59], while the latter shows that purification can be incorporated into it using optimistic (blind) execution [17], [18]. That feasibility result addresses whether purification can be integrated, but not how it should be scheduled; deciding where and how often to purify across a full chain is left open, which Section 2.5 returns to.

2.4 Schedule Optimisation in Quantum Networks

Deciding when and where to apply entanglement purification is a separate problem, even once purification is available at every stage of a repeater chain. In adaptive schemes, a repeater node chooses its next operation based on fidelity or heralding outcomes observed so far, rather than following a fixed plan [60]. Reinforcement learning and other data-driven controllers have also been proposed for related quantum-network control problems. For example, one study reports substantial offline costs in richer settings, including state representations that grow exponentially, training times of up to ~ 36 hours per configuration, and hyperparameter searches required to outperform a heuristic baseline [61]. Optimistic purification has also been studied directly in generic quantum networks, independently of any RGS-specific construction [18], showing that deferring heralding can be useful beyond the architecture studied here. The choice of how to multiplex the qubits of a logical channel across serial or parallel physical links is another scheduling decision, with a measurable impact on end-to-end rate and fidelity [62].

2.5 Positioning of This Work

The work surveyed here either optimises a fixed component of a repeater design, such as a purification circuit or graph-state construction, or evaluates a specified scheduling policy. For the specific HRGS architecture studied in this thesis, no systematic search considers where to purify, how many copies to consume, which circuit to use, and in what order across a full chain. As Section 1.2 establishes, this is the gap addressed by this thesis.

Chapter 3

Background and Preliminaries

This chapter introduces the physical operations and mathematical quantities that Chapter 4 assembles into a searchable schedule model.

Section 3.1 reviews Bell pairs and the HRGS repeater architecture proposed in the Bridging paper, whose fixed protocol-level operations form what Chapter 4 calls the *backbone layer*. Section 3.2 introduces the two quantities used to track a resource’s quality through a schedule: the error vector and the Z side-effect parity, together with the decoherence law that acts on the former while a resource is idle. Section 3.3 then covers the Integrating paper’s purification circuits and the distinction between heralded and optimistic execution, which determines their timing cost.

3.1 Quantum Networks and Repeaters

3.1.1 Bell Pairs and Fidelity

A Bell pair is an entangled two-qubit state shared between two nodes. The canonical Bell state is

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Its quality is measured by a single scalar, the fidelity $F = \langle \Phi^+ | \rho | \Phi^+ \rangle$, which quantifies how close the shared state ρ is to a perfect pair ($F_{\max} = 1$). Depolarising noise on the transmitted qubit and memory decoherence reduce F ; photon loss instead lowers the probability that a pair can be established in the first place. Chapter 4 evaluates each finished schedule by its fidelity, rate, and resource cost for producing such a pair between the end nodes. Improving one of these quantities is meaningful only relative to this definition of F .

While Chapter 2 surveys why direct photonic transmission cannot distribute such pairs over long distances, we introduce here the elementary operation used to establish and extend links: a two-qubit **Bell-State Measurement (BSM)**. Realized with passive linear optics and no ancillary photons, a BSM succeeds with probability at most 50% per attempt, conditional on the photons arriving [16], [42]. This ceiling is why the architecture below builds redundancy into every hop.

3.1.2 Half-RGS Repeater Architecture

A *graph state* associates one qubit with every vertex of a graph $\mathcal{G} = (V, E)$: each qubit starts in state $|+\rangle$ and a controlled- Z gate is applied along every edge of the graph [32]–[34], [38]. The result is a multipartite entangled resource whose measurement outcomes at one vertex can be used, with appropriate local single-qubit operations, to establish entanglement between other vertices [35], [36], [38]. This is the underlying resource of the architectures surveyed in Section 2.3, including the one this thesis builds on.

A *repeater graph state* (RGS) is one such graph state, with $2m$ inner qubits and $2m$ outer qubits, where m is the number of outer-qubit arms on each side. It is proposed as the basis for a memory-free all-photonic quantum repeater [15]. The inner qubits form a complete bipartite graph and are tree-encoded [48], allowing the logical measurements required by the protocol to be performed despite photon loss [50]. Each outer qubit is attached to one inner qubit and is consumed by a BSM with a neighbouring station. Such a network has two station types: **Repeater Graph State Source (RGSS)** nodes, which generate the RGS and hold it only until it is transmitted, and **Advanced Bell-State Analyser (ABSA)** nodes, which perform the BSMs between neighbouring stations and forward only the resulting entanglement, never generating anything themselves [15], [16].

The Bridging paper’s own contribution is the **HRGS**: half of an RGS, consisting of an *anchor qubit* together with its associated inner and outer qubits, generated and held at a single station rather than requiring the full $2m$ -qubit state to exist in one place [16]. Two HRGSs are combined into a full RGS by a controlled-phase gate followed by an XX measurement on their two anchor qubits, which realizes entanglement swapping and is treated as a single operation throughout this thesis (Section 3.1.3). Because a HRGS only needs to be generated and briefly held, an end node can act as its own source: it generates a HRGS using its stationary anchor and emits the photonic qubits toward its neighbouring ABSA, rather than requiring a separate RGSS. This is what lets end-node memory scale additively with the number of hops instead of multiplicatively, the central resource-saving claim of the Bridging paper.

An N -hop chain between two end nodes has $N + 1$ stations numbered $0, \dots, N$: stations 0 and N are the end nodes (Alice and Bob), and stations $1, \dots, N - 1$ are RGSS nodes. Every RGSS generates a full RGS and distributes its two halves to the ABSA on each side; an end node generates only the one HRGS it needs and sends its photonic qubits to its single neighbouring ABSA. Each hop i , the link between stations $i - 1$ and i , has one ABSA that receives outer qubits from both sides and performs the requisite BSMs, so that entanglement is established one hop at a time until it spans the whole chain [16].

3.1.3 Backbone Operations and Spans

Because the BSM’s success probability is capped at 50% (Section 3.1.1), each hop provisions k_i redundant outer-qubit arms rather than a single pair; Chapter 4 uses k_i as the per-hop arm-count parameter. At every ABSA, once one arm’s BSM succeeds, that arm’s inner qubits are *kept*: the station *discards* the redundancy by measuring every other arm’s inner qubits in the Z basis [15], [16]. This *backbone selection* is forced by the protocol at every hop, on every trial.

A resource that has not yet crossed any hop – a freshly generated HRGS anchor or two such

anchors joined at the same station, for instance – is still *local* to its station of origin. Once a chain of **BSMs** and join operations connects stations across one or more hops, it forms a resource connecting stations a and b . Its *span* is written $\text{Span}(a, b)$, or simply (a, b) where no ambiguity arises. The symbol κ denotes the current stage of a resource, either local to a single station or spanning some interval of the chain; Chapter 4 gives this notation a full formal treatment. Two properties of spans matter for everything that follows:

1. A resource’s span is always a contiguous interval of stations, so that if it spans (a, b) then it also spans every intermediate station $a < i < b$. This is because every backbone operation that extends a span (a **BSM** or a join) connects resources across adjacent stations or at a shared endpoint.
2. A resource’s span is independent of its history. Two resources that have the same span (a, b) may have reached that state by entirely different routes, e.g. one purified at every hop and the other not at all, since all that matters is the current extent of the span.

Every **BSM**, arm measurement, and join operation also produces a possible Pauli- Z side effect on the resulting resource, which must eventually be corrected. This side effect is what Section 3.2.2 tracks formally; the key fact needed here is that resolving it costs only a small, constant amount of classical communication per **ABSA** per trial, regardless of how deep the tree-encoded inner qubits are, since each arm’s side effects collapse to a couple of bits before the **ABSA** passes them on [16].

3.2 Error Representation

3.2.1 The Error Vector

Just like the two source papers [16], [17], this thesis models outer-photon depolarising noise, tree-encoded inner-qubit measurement error, and memory decoherence (Section 3.2.3) as Pauli channels acting on an otherwise-ideal Bell pair. Conditional on a successful transmission, the resulting state remains diagonal in the Bell basis at every stage. It is therefore represented by a real, normalized vector of four probabilities,

$$\mathbf{e} = (w, x, y, z)^\top, \quad w + x + y + z = 1, \quad (3.1)$$

corresponding to the Bell-diagonal density operator

$$\rho(\mathbf{e}) := w |\phi_{II}\rangle\langle\phi_{II}| + x |\phi_{ZI}\rangle\langle\phi_{ZI}| + y |\phi_{ZZ}\rangle\langle\phi_{ZZ}| + z |\phi_{IZ}\rangle\langle\phi_{IZ}|. \quad (3.2)$$

The four components are the probabilities of no error, a Z error on the first qubit only, a Z error on both qubits, and a Z error on the second qubit only, respectively [17]. The first component w is exactly the fidelity of Section 3.1.1 and it is the quantity Chapter 4 reads off as the fidelity cost function $F(\Sigma; \mathcal{N})$. The same formalism describes a pre-transmission local pairing (an anchor and its own outer qubit, before either has crossed a hop) just as it does an established edge, which is what allows purification to be defined once and reused identically at

every stage of the schedule (Section 3.3.2). This representation is closed under every operation the backbone and scheduling layers perform.

At the error-vector level, a **BSM** and a join operation apply the same bilinear composition rule to their two inputs:

$$\text{BSM}(\mathbf{e}_1, \mathbf{e}_2) = \begin{bmatrix} w_1 w_2 + x_1 x_2 + y_1 y_2 + z_1 z_2 \\ w_1 x_2 + x_1 w_2 + z_1 y_2 + y_1 z_2 \\ w_1 y_2 + y_1 w_2 + x_1 z_2 + z_1 x_2 \\ w_1 z_2 + z_1 w_2 + x_1 y_2 + y_1 x_2 \end{bmatrix}, \quad (3.3)$$

which again yields a Bell-diagonal vector [16], [17]; purification (Section 3.3.2) and idle-time decoherence (Section 3.2.3) preserve the same property. Two independent contributions feed into $\mathbf{e}$ as a resource crosses a hop: an outer-qubit term from the depolarising rate e_d applied to the transmitted photon and an inner-qubit term from the tree-encoded measurement error rates $p_{\text{in}}^X, p_{\text{in}}^Z$, which contributes only to the independent x and z components, never to y [16]. Under otherwise-isotropic noise, this biases the accumulated error toward independent single-sided Z errors (x, z) over correlated two-sided ones (y), a bias that Section 3.3.1 shows the purification circuits can exploit.

3.2.2 Z Side-Effect Algebra

Independently of the error vector, every backbone operation (Section 3.1.3) may flip the sign of the surviving qubit's stabilizers, a Pauli- Z *side effect*. Unlike $\mathbf{e}$, a side effect is exactly known once its source measurement outcome is known; it therefore needs only a single bit, $s \in \mathbb{F}_2$ per resource, rather than a probability distribution. Side effects compose by **eXclusive OR (XOR)** whenever two resources are merged at a **BSM**, an arm measurement, or a join. This composition rule is exact, since a Pauli- Z commutes through every other operation in the model up to a sign that is tracked the same way [16]. The parity accumulated by a **HRGS** during its own generation, from the sequence of emitter operations used to build its tree-encoded inner qubits, is folded into this same bit before the **HRGS** is ever used, so no separate bookkeeping is needed for it. At the very end of a schedule, the total accumulated parity s^{total} at an end node is the **XOR** of every contributing side effect and a single conditional correction $Z^{s^{\text{total}}}$ applied at that point is enough to recover the intended state.

This parity bookkeeping is separate from the error vector: s determines the Pauli frame in which a resource is expressed, while $\mathbf{e}$ determines its noise. Since all side effects within an arm reduce to their **XOR** parity, each arm contributes only the constant-size classical message noted in Section 3.1.3, regardless of the depth of its tree encoding [16]. The scheduler tracks the parity for correctness, but compares candidate schedules through the error vector and the cost functions derived from it.

3.2.3 Decoherence Model

While idle, a resource’s error vector relaxes toward the maximally mixed state at a rate set by the memory’s coherence time, $T_2 = 1/\gamma$, the standard coherence-time convention [63]:

$$\mathbf{e}(t) = \frac{1}{4}\mathbf{1} + e^{-\gamma t}(\mathbf{e}(t_0) - \frac{1}{4}\mathbf{1}).$$

This relaxation is applied locally and independently at every node of a schedule to the exact idle interval that node experiences, rather than just once as a single correction at the end. Consequently, two resources that wait for different durations must be decohered before they are combined; replacing those waits by one averaged duration after composition would generally produce a different error vector [16], [17].

How much idle time a resource accumulates depends entirely on how a schedule is arranged, which is why the Integrating paper [17] singles out three canonical timing patterns that later serve, in Chapter 6, as a validation target for this thesis’ own cost functions:

- *Raw*: a chain with no purification at all, so every resource is used as soon as it is generated.
- *Baseline*: each purification round is heralded before the next round begins. This avoids speculative purification but incurs a separate classical confirmation delay after every round.
- *Optimistic*: generation and purification proceed back-to-back, with intermediate confirmation deferred until the end of the sequence. Section 3.3.3 explains the resulting communication-time saving and its effect on rate.

3.3 Entanglement Purification

3.3.1 Purification Circuits: $\mathcal{C}_{YY}$, $\mathcal{C}_{ZX}$, and $\mathcal{C}_{XZ}$

Purification trades copies for quality, as surveyed in Section 2.2. In this architecture, a circuit acts on two resources at the same span κ (Section 3.1.3) through a probabilistic LOCC measurement. On success, it consumes the two inputs and leaves one output resource with an updated error vector; on failure, both inputs are discarded. The Integrating paper adapts this idea to the RGS-based resources used here [17]. Three such circuits are available, distinguished by the stabilizer they jointly measure across the two input pairs: $\mathcal{C}_{ZX}$ detects a Z error on the second qubit of the pair, $\mathcal{C}_{XZ}$ detects a Z error on the first qubit [14], and $\mathcal{C}_{YY}$ detects the corresponding parity of Z errors across the four qubits involved [17]. Each circuit is heralded by comparing the two parties’ local measurement outcomes; Section 3.3.3 distinguishes immediate from deferred confirmation. Because the inner-qubit noise term biases accumulated error toward independent single-sided errors (x, z) rather than correlated ones (y), purification sequences can exploit this bias by applying $\mathcal{C}_{YY}$ first, before switching to $\mathcal{C}_{ZX}$ or $\mathcal{C}_{XZ}$ (Section 3.2.1).

3.3.2 Success Probabilities and Output States

The success probability and purified output vector for each circuit [17] are:

$$P_{C_{ZX}}(\mathbf{e}_1, \mathbf{e}_2) = (w_1 + x_1)(w_2 + x_2) + (z_1 + y_1)(z_2 + y_2), \quad (3.4)$$

$$P_{C_{XZ}}(\mathbf{e}_1, \mathbf{e}_2) = (w_1 + z_1)(w_2 + z_2) + (x_1 + y_1)(x_2 + y_2), \quad (3.5)$$

$$P_{C_{YY}}(\mathbf{e}_1, \mathbf{e}_2) = (w_1 + y_1)(w_2 + y_2) + (x_1 + z_1)(x_2 + z_2). \quad (3.6)$$

$$\text{Pur}_{C_{ZX}}(\mathbf{e}_1, \mathbf{e}_2) = \frac{1}{P_{C_{ZX}}} \begin{bmatrix} w_1 w_2 + z_1 z_2 \\ z_1 w_2 + w_1 z_2 \\ x_1 y_2 + y_1 x_2 \\ x_1 x_2 + y_1 y_2 \end{bmatrix}, \quad (3.7)$$

$$\text{Pur}_{C_{XZ}}(\mathbf{e}_1, \mathbf{e}_2) = \frac{1}{P_{C_{XZ}}} \begin{bmatrix} w_1 w_2 + x_1 x_2 \\ z_1 z_2 + y_1 y_2 \\ z_1 y_2 + y_1 z_2 \\ x_1 w_2 + w_1 x_2 \end{bmatrix}, \quad (3.8)$$

$$\text{Pur}_{C_{YY}}(\mathbf{e}_1, \mathbf{e}_2) = \frac{1}{P_{C_{YY}}} \begin{bmatrix} w_1 w_2 + y_1 y_2 \\ x_1 z_2 + z_1 x_2 \\ y_1 w_2 + w_1 y_2 \\ x_1 x_2 + z_1 z_2 \end{bmatrix}. \quad (3.9)$$

3.3.3 Heralded versus Optimistic Purification

In the *heralded* model, each purification round waits for a full classical round-trip ($2L/c$, where L is the end-to-end fiber length and c is the signal propagation speed) to confirm the previous round's success before the next round begins. The *optimistic* (or blind) model [17], [18] instead runs generation and purification back-to-back without intermediate confirmation: operations are executed speculatively, and their classical outcomes are compared only in a deferred communication phase at the end. This replaces the n_{pur} round-trip heraldings that heralded execution would otherwise require with a single final heralding phase, which is the main source of the rate advantage reported in [17].

Crucially, *optimistic* is not a property of an individual purification operation. It follows from the placement of heralding steps relative to purification steps in the complete schedule. Section 4.3 states this criterion precisely in the formal DAG representation.

Chapter 4

The Formal Model

As established in Chapter 1 (Section 1.2), the Integrating paper [17] demonstrates purification schedules that mix generation, link-level, and end-node purification through a single hand-picked example, leaving the search for a better schedule as an open problem. Turning this into a problem that can actually be searched requires a precise object to search over. This chapter builds that object step by step: the given physical network configuration (Section 4.1), the state carried by each partially built Bell pair (Section 4.2), the schedule itself as a graph of operations (Section 4.3), the resources it is allowed to consume (Section 4.4), and finally the optimization problem this thesis addresses (Section 4.5). Section 4.6 closes the chapter by stating precisely which schedules are excluded from this model and why.

The building blocks introduced in Chapter 3 – generation, joining, and backbone selection – are reused without redefinition below. Together, they form the *backbone layer*: the fixed, protocol-determined process of establishing entanglement. This chapter defines the *scheduling layer* built on top of it: where to insert purification, which circuit to use, in what order to apply operations, and how to compare the resulting schedules.

4.1 Network Configuration

A network configuration fixes the physical parameters that a schedule cannot change. Stations are numbered $0, 1, \dots, N$ along the chain, so a hop is the link between consecutive stations $i - 1$ and i : an N -hop chain has N hops and $N + 1$ stations. With $i = 1, \dots, N$ indexing the hops, the configuration is the tuple

$$\mathcal{N} = \left(N, \{ \ell_i \}, \{ \vec{b}^{(i)} \}, \{ k_i \}, \{ (p_{\text{in},i}^X, p_{\text{in},i}^Z) \}, \{ \eta_i \}, e_d, \gamma, c, \tau_{\text{emit}} \right),$$

where, for each hop i , ℓ_i is its length; $\vec{b}^{(i)}$ is the list of tree-encoding branching factors at successive levels; k_i is its number of redundant BSM arms; $p_{\text{in},i}^X$ and $p_{\text{in},i}^Z$ are its inner-qubit measurement error rates; and η_i is its photon survival probability. The remaining parameters are shared across the chain: e_d is the outer-photon depolarizing rate, γ is the memory dephasing rate, c is the signal propagation speed in the fiber, and τ_{emit} is an optional per-photon emission time used to model branching-dependent generation latency. Conditional on photon arrival, the per-arm BSM success probability is capped at 50% [16], [42]. The factor η_i accounts separately for transmission

loss, while k_i supplies redundancy against the measurement ceiling.

Unless stated otherwise, experiments start from the Integrating paper configuration, written $\mathcal{N}_{\text{ref}}$. It has identically configured hops:

$$N = 10, \quad \ell_i = 2 \text{ km}, \quad \vec{b}^{(i)} = (16, 14, 1), \quad k_i = 18 \quad \forall i, \quad e_d = 0.01, \quad \gamma = 0,$$

$$p_{\text{in},i}^X = p_{\text{in},i}^Z = 0 \quad \forall i, \quad c = 2 \times 10^5 \text{ km/s}.$$

The remaining per-hop parameter, η_i , is computed from ℓ_i using attenuation $\alpha = 0.2 \text{ dB/km}$, giving $\eta_i = 10^{-\alpha \ell_i / 10} \approx 0.912$. [depends on Whit3z’s answer to timing question on slack: The reference configuration leaves τ_{emit} unset, so the default evaluator adds no branching-dependent generation offset to the latency]. Section 6.8 explicitly enables and varies this parameter.

Table 4.1: Reference configuration and parameter studies. Values marked “held fixed” remain unchanged in the reported sweeps.

Parameter	Description	Reference value	Varied in
N	Number of hops	10	Section 6.7
ℓ_i	Hop length	2 km	held fixed
$\vec{b}^{(i)}$	Branching vector	(16, 14, 1)	held fixed
k_i	Number of redundant BSM arms	18	Section 6.6
$p_{\text{in},i}^X, p_{\text{in},i}^Z$	Inner-qubit measurement error rates	0, 0	Section 6.6
η_i	Photon survival probability	≈ 0.912 (derived)	held fixed
e_d	Outer-photon depolarizing rate	0.01	Sections 6.4, 6.7.1
γ	Memory dephasing rate	0	Section 6.8
τ_{emit}	Per-qubit emission or gate-cycle time	unset	Section 6.8
c	Signal propagation speed in the fiber	$2 \times 10^5 \text{ km/s}$	held fixed

4.2 Representing a Partially-Built Bell Pair

Every resource produced while a schedule is evaluated, from a local HRGS to the final end-to-end pair, is represented by a state:

$$S = (\mathbf{e}, s, t, t_{\text{gen}}, \kappa, r, h).$$

The vector $\mathbf{e}$ is the error vector introduced in Section 3.2.1, and s is the Z side-effect parity from Section 3.2.2. The remaining fields are introduced by the scheduling layer:

- t and t_{gen} are the current simulated time and the time at which the resource was first generated, respectively. Their difference, $\Delta t_{\text{idle}} = t - t_{\text{gen}}$, records the elapsed waiting time relevant to the decoherence model of Section 3.2.3.
- $\kappa \in \mathcal{K}$ is the *span* of the resource, where $\mathcal{K} = \{\text{RGSS}\} \cup \{\text{Span}(a, b) : 0 \leq a < b \leq N\}$ is the set of possible stages for a given network configuration (defined formally in Section 4.3): either RGSS for a resource that has not yet crossed a hop, or $\text{Span}(a, b)$ for an edge that already connects station a to station b .
- r counts how many purification rounds have already been applied to this branch.

- $h \in \{\textit{pending}, \textit{resolved}\}$ records whether the classical heralding signal for this resource has been received yet.

The field that matters most for what a schedule is allowed to do is κ : two states can only be combined by a purification circuit if they carry the *same* κ . Two branches with different histories, one purified twice and the other once for instance, can still be combined as long as both currently have the same span as previously stated in Section 3.1.3. This rule lets the model express schedules that mix purification strategies across different parts of the chain, including the type of schedule illustrated in the source paper’s Fig. 4 [17].

4.3 Schedules as Directed Acyclic Graphs

A schedule is written as a pair $\Sigma = (T, \phi)$, where T is a rooted **Directed Acyclic Graph (DAG)** and ϕ is a labeling function

$$\phi : T \rightarrow \mathcal{K}, \quad \mathcal{K} = \{\text{RGSS}\} \cup \{\text{Span}(a, b) : 0 \leq a < b \leq N\},$$

assigning to every node of T the span $\kappa \in \mathcal{K}$ of the state that node produces, using the same κ introduced in Section 4.2. Each node kind represents a distinct physical or logical action. The colours and shapes below form the legend used in the worked schedule of Figure 4.1:

-  **GenNode** (light-blue ellipse): **HRGS** generation; these nodes are always leaves of T at $\kappa = \text{RGSS}$;
-  **AbsaBsmNode** (orange box): the photonic **BSM** that combines two **RGSS**-local resources associated with one hop into a single-hop edge;
-  **JoinNode** (green box): entanglement swapping, stitching two compatible spans into a wider one;
-  **PurifyNode** (purple box): a two-to-one $\mathcal{C}_{YY}$, $\mathcal{C}_{ZX}$, or $\mathcal{C}_{XZ}$ circuit (Section 3.3.1);
-  **HeraldNode** (yellow diamond): resolving the classical heralding signal for a branch;
-  **IdleNode** (grey dashed box): an explicit wait, used only where the schedule’s timing requires one;
-  **PauliCorrectNode** (red double circle): the root of T at $\kappa = \text{Span}(0, N)$, representing the finished end-to-end pair after final Pauli correction.

Optimistic versus heralded schedules A schedule Σ is *fully optimistic* when no **HeraldNode** separates consecutive **PurifyNode** steps on any branch. In the optimistic schedules used here, all heralding is deferred to one **HeraldNode** directly below **PauliCorrectNode** at the root. A schedule is *fully heralded* when a **HeraldNode** is placed between consecutive purification rounds; each intermediate herald forces a classical round-trip ($2L/c$) before the next round may begin. Optimism is therefore a property of the schedule’s timing structure, represented here by the placement of **HeraldNodes**, rather than a property of an individual purification operation. The

DAG representation can express mixed timing patterns, but the experiments compare only fully optimistic and fully heralded schedules.

4.3.1 Legality Conditions

A graph T is a *structurally legal* schedule when it has one **PauliCorrectNode** root, is acyclic, contains no unreachable nodes, and obeys the following stage constraints:

- (i) **GenNodes** may only appear at $\kappa = \text{RGSS}$;
- (ii) an **AbsaBsmNode** consumes two **RGSS**-local resources associated with one hop and produces $\text{Span}(i, i + 1)$;
- (iii) a **PurifyNode**'s two children must share the same κ , i.e. the span-matching rule stated informally in Section 4.2;
- (iv) a **JoinNode**'s two children must be either two **RGSS**-local resources or spans that share an endpoint. For the latter, i.e. (a, b) and (b, c) for some $a < b < c$, the two spans need not come from neighbouring hops, only meet at station b , and the resulting resource has span (a, c) ;
- (v) **IdleNode** and **HeraldNode** [to simplify maybe?: are single-child pass-through operations that copy their child's κ unchanged, since neither an explicit wait nor resolving a heralding signal changes which stations a resource spans];
- (vi) the root must receive a state with $h = \text{resolved}$ (Section 4.2) at $\kappa = \text{Span}(0, N)$.

The implementation checks the root, graph structure, and stage consistency before evaluation, while the evaluator repeats the physical preconditions on the corresponding resource states.

This representation was checked against the source paper's own Fig. 4 schedule [17], which builds one Bell pair from purified individual links, a second from two purified 5-hop segments, and a third with no purification at all, then combines all three with a final purification round at the end nodes. It combines branches purified to different degrees and built up in different ways, and this is exactly the class of schedule this model can express: allowing κ to be an arbitrary span, and requiring **PurifyNode** to check only κ and not how each branch was built, is what makes that combination legal.

Section 6.2 confirms that this representation is not merely descriptive: evaluating the raw, end-node-heralded, and end-node-optimistic canonical schedules through the same bottom-up pass used elsewhere in this thesis reproduces the source paper's fidelity values to four significant figures.

4.4 Resource Budget

A schedule is only meaningful together with a budget

$$B = (n_{\text{pur}}, e_{\text{max}}, M_{\text{max}}),$$

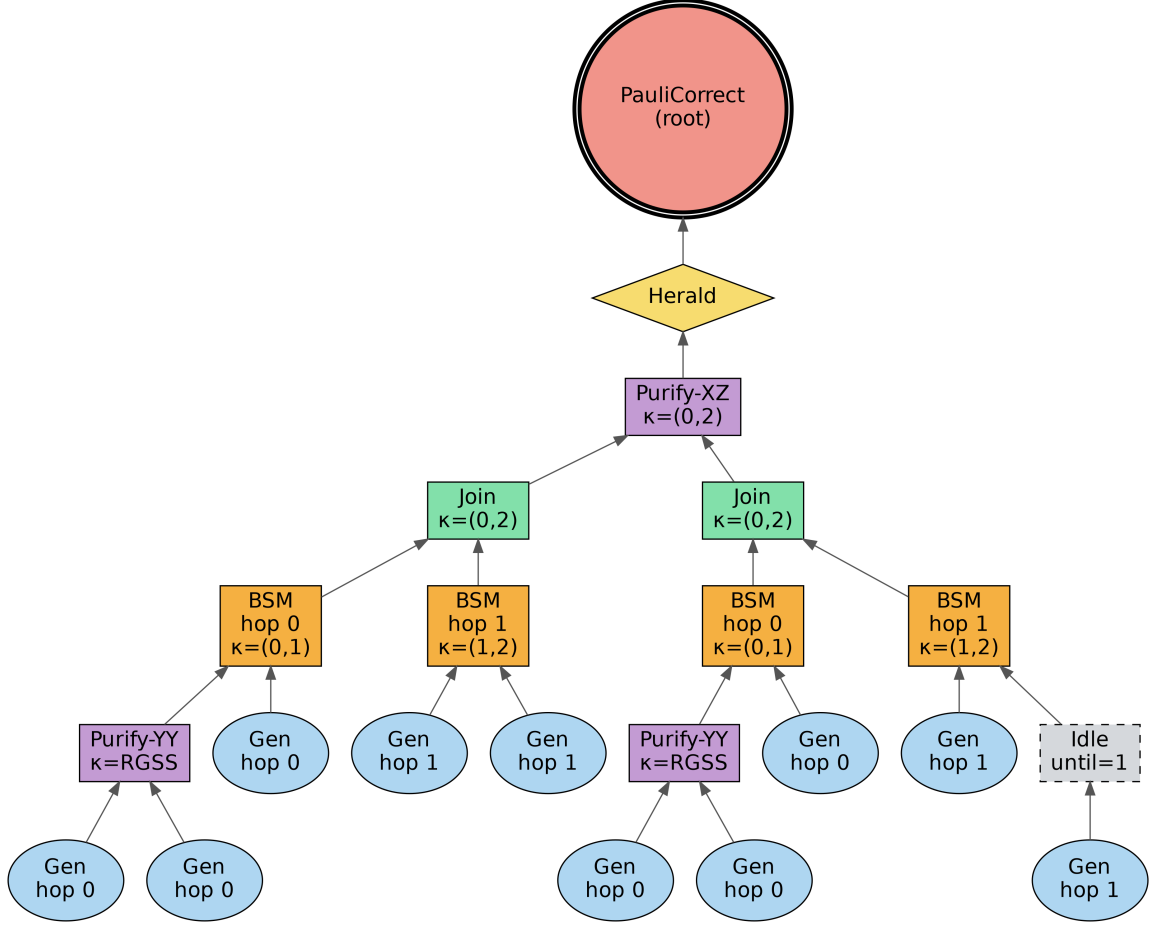


Figure 4.1: $N = 2$ example schedule Σ illustrating all seven node types ($C(\Sigma) = 10$ GenNode leaves). **Copy A** (left subtree): hop 0 applies PurifyNode-YY at $\kappa = \text{RGSS}$ before the outer-photon AbsaBsmNode; hop 1 is raw. **Copy B** (right subtree): identical except the right-side hop-1 GenNode is wrapped in an IdleNode (grey, dashed) modelling a synchronisation wait before the AbsaBsmNode. The two copies are joined into $\kappa = (0, 2)$ by a JoinNode, then combined by PurifyNode-XZ. The single HeraldNode placed after (not before) PurifyNode-XZ makes this an optimistic schedule (Section 4.3). PauliCorrectNode is the root at $\kappa = (0, 2)$. Edges run bottom-to-top.

where n_{pur} specifies the number of extra HRGS copies generated for each purification round, e_{max} is the maximum number of Gen leaves T is allowed to contain, and M_{max} is the maximum number of branches that may be under construction at the same time.

A schedule Σ is *formally feasible* with respect to B if it uses at most e_{max} Gen leaves and never holds more than M_{max} branches open at once. The resource cost $C(\Sigma)$ used throughout Chapter 6 is simply the number of Gen leaves in T , and every headline result enforces $C(\Sigma) \leq e_{\text{max}}$. M_{max} is enforced separately: since T only fixes a topological partial order on its operations, not a concrete real-time execution order, the number of branches simultaneously open depends on which legal execution order a scheduler chooses. For the tree-shaped candidate schedules considered here, the implementation computes $M(\Sigma)$, the minimum peak number of concurrently open branches over every legal execution order, via the classical Sethi–Ullman register-allocation recurrence for expression trees [64]: a Gen leaf needs one register; IdleNode/HeraldNode inherit their child’s register count because they preserve the same physical resource; and a 2-input node (Join/AbsaBsm/Purify) with children needing L and R registers needs $\max(L, R)$ if $L \neq R$,

or $L + 1$ if $L = R$ (the classical tie case, where the first-finished side’s result must be held while the second side is built). This recurrence gives the minimum register requirement for a binary expression tree, so $M(\Sigma)$ is exact for the tree-shaped schedules considered here and is computed in a single $O(|T|)$ bottom-up pass without timing information. Unlike F , R , and L , $M(\Sigma)$ depends only on the schedule structure. Σ is feasible with respect to $M_{\max}$ if and only if $M(\Sigma) \leq M_{\max}$, checked directly by the **DAG**-level method and, where requested, used as a constraint on the search objective (Section 4.5); Section 4.6 discusses the resulting scope limitation.

The source paper’s own budget convention is $e_{\max} = 10N$: five **HRGS** copies per side, on both sides, for every hop. This convention is used as the reference budget throughout the experiments, and Section 6.7 asks directly whether it remains large enough as N grows.

4.5 Objective Functions and the Optimization Problem

Four metrics can be read from the finished schedule **DAG** by a single bottom-up pass:

- fidelity $F(\Sigma)$, the w component of the error vector at the root;
- rate $R(\Sigma) = P_{\text{success}}/T_{\text{attempt}}$, where T_{attempt} is the deterministic wall-clock duration of one complete schedule attempt. Under the renewal model, a failed non-adaptive purification attempt restarts the entire schedule, so the expected wall-clock time per successful pair is $T_{\text{attempt}}/P_{\text{success}}$, whose reciprocal is the stated rate;
- resource cost $C(\Sigma)$, defined in Section 4.4;
- latency $L(\Sigma)$, the makespan of T .

The optimization problem this thesis addresses is then, given a resource budget B (Section 4.4),

$$\Sigma^*(\mathcal{N}, B) = \underset{\substack{\Sigma \text{ legal w.r.t. } \mathcal{N} \\ \Sigma \text{ feasible w.r.t. } B}}{\arg \max} R(\Sigma; \mathcal{N}) \quad \text{s.t.} \quad F(\Sigma; \mathcal{N}) \geq F_{\min}.$$

In other words, among the legal schedules that are feasible with respect to B (Section 4.4: use at most $e_{\max}$ **Gen** leaves and never hold more than $M_{\max}$ branches open at once), find the one with the highest rate that still clears a minimum fidelity requirement. This is the objective used in every headline result in Chapter 6 unless stated otherwise, with $F_{\min} = 0.9$ matching the source paper’s own threshold and $M_{\max} = \infty$ (no concurrency cap) unless a specific experiment states otherwise (Section 4.6).

The same four metrics support other research questions by swapping which one is maximized and which are constrained: maximizing F subject to a rate floor asks for the best achievable quality at a given throughput; minimizing C subject to floors on F and R asks for the cheapest schedule that still meets both requirements. Both variants were implemented as alternative objective presets while exploring these questions. Section 6.5.1 shows that switching between all three presets requires no change to the search algorithms of Chapter 5: only the scoring function and feasibility constraints need to be swapped.

4.6 Scope and Assumptions

This model rests on three simplifying choices, stated explicitly here before any algorithm or result depends on them.

First, only *non-adaptive* schedules are represented: T is fixed before execution starts and cannot branch on an intermediate measurement or heralding outcome. A fully adaptive scheme would require a different formulation, most naturally a [Markov Decision Process \(MDP\)](#) [65], [66]. This is left to future work (Section 8.2).

Second, $M_{\max}$ is computed exactly for any concrete Σ (Section 4.4) and can be enforced as a feasibility constraint on the search objective, but this constraint is only applied to whole, already-built candidate schedules at the point they are scored – it is not yet threaded into the recursive span-partition search of Chapter 5 as an early-pruning criterion in the way $e_{\max}$ is.

The reason is that $M(\Sigma)$, unlike $C(\Sigma)$, is not additive across independently searched sub-spans: two sub-schedules built for disjoint spans can be scheduled concurrently or sequentially, so pruning on $M_{\max}$ during search would require tracking register-count information across sibling spans, not just within each span’s own subtree, risking the removal of a candidate that is only infeasible under a poor execution order. Every reported headline result therefore still optimizes without an $M_{\max}$ cap; where a cap is applied (Section 6.7), it changes which already-generated candidate is selected as best, not how many candidates are generated in the first place.

Third, conditional on a successful transmission, the fidelity-noise model throughout this thesis is a Pauli channel [63], [67], [68], following the assumption made by the source papers; photon loss instead reduces the success probability P_{success} rather than the fidelity F . Comparisons between schedules are therefore made under this specific fidelity-noise model, so a ranking that holds here is not automatically guaranteed to transfer to a device dominated by a different error channel. This point is returned to in Section 7.6.

Chapter 5

Search Methods

Three search tiers are implemented on top of one shared schedule representation and one shared evaluator. Every returned candidate is a [or valid?: concrete] schedule Σ : it passes the structural checks of Section 4.3.1 and is evaluated by the bottom-up pass from Section 4.5. The tiers differ only in which legal DAGs they consider. Search approximations can exclude a promising DAG before final ranking, but they do not change how fidelity, success probability, latency, or cost is calculated for a DAG that is returned.

The chapter moves from general search concepts to evaluation and then to increasingly broad but less exhaustive search. Section 5.1 introduces the three combinatorial-optimisation ideas used throughout the chapter. Section 5.2 describes the shared evaluator. Section 5.3 introduces the fixed baseline families, which are also retained as a safeguard against frontier pruning. Section 5.4 presents the span-partition recurrence, Section 5.5 introduces the additional same-span pumping operation and its search cap, and Section 5.6 describes the beam-search heuristic used for the production-scale experiments in Chapter 6.

5.1 Preliminaries: Combinatorial Search Concepts

The schedule space Σ ranges over the legal DAGs defined in Section 4.3 and is combinatorially large: even a modest chain admits combinatorially many legal schedules, far too many to enumerate one at a time. The three search tiers below borrow standard tools from combinatorial optimisation to make this space tractable. This section introduces each tool in general terms before it is applied to schedules specifically; a reader already familiar with Pareto optimality, dynamic programming, and beam search may skip ahead to Section 5.2.

Multi-objective optimisation and Pareto dominance A search problem is *multi-objective* when candidates are compared along several quantities at once rather than a single score, here cost C , fidelity F , and success probability P_{success} . Candidate A *Pareto-dominates* candidate B if A is at least as good as B in every quantity and strictly better in at least one; a candidate is *Pareto-optimal* (non-dominated) if no other candidate dominates it, and the *Pareto frontier* is the set of all non-dominated candidates [69]–[71]. Keeping the whole frontier, rather than a single best-scoring candidate, matters whenever a locally inferior choice can become the better choice after it is combined with another resource later, as occurs when sub-spans are joined.

Dynamic programming *Dynamic programming* solves a large problem by decomposing it into smaller and overlapping subproblems that can be solved once and reused wherever they recur (*memoized*) [72]. Section 5.4 applies this idea to schedules: candidate sub-schedules for a short span are computed once and reused when constructing longer spans.

Heuristic and beam search When the number of subproblems or candidates becomes too large for exhaustive enumeration, a *heuristic search* trades completeness for tractability by discarding some candidates before they are fully explored. *Beam search* is one such heuristic: at each stage it retains only a fixed number of candidates, the *beam width*, according to a chosen ranking rule [73]–[75]. A narrower beam reduces computational cost but increases the chance of discarding a candidate that would have led to a better final schedule; increasing the beam width retains more of the search space at greater cost. Section 5.6 applies this principle to the span-partition recursion of Section 5.4.

5.2 The Schedule Evaluator

The evaluator makes one bottom-up pass over a schedule DAG. At each node it applies the corresponding physical operation to the current Pauli-channel state, synchronizes branches that become ready at different times, applies the resulting memory decoherence, and propagates side effects and timing. At the root it returns fidelity F , success probability P_{success} , resource cost C , and latency L . Rate is then $R = P_{\text{success}}/L$ under the full-restart model of Section 4.5.

Section 6.2 shows that this pass reproduces the source paper’s Fig. 5 fidelity numbers to four significant figures. That agreement validates the physical model implemented by the evaluator; it does not establish that any search tier finds an optimal schedule, which is a separate question addressed by the remainder of this chapter and by Chapter 6.

5.3 Fixed Schedule Families

The first tier enumerates four fixed structural families: (1) a single raw end-to-end chain, with no purification; (2) end-node pumping, in which n_{pur} independent raw chains are purified sequentially at the end nodes with a round-trip herald between rounds; (3) the optimistic counterpart, which defers classical heralding until the end; and (4) uniform link-level purification, applying the same copy count and circuit sequence at every hop. For even N within budget, the Integrating paper’s hand-chosen schedule [17] is included as a fifth fixed comparison point.

Within each family, purification sequences use the three two-copy circuits $\mathcal{C}_{YY}$, $\mathcal{C}_{ZX}$, and $\mathcal{C}_{XZ}$ (Section 3.3.1). All 3^r circuit sequences are enumerated up to a configured round cutoff r_{max} ; above that cutoff, the implementation uses a small curated set. Thus, “brute force” denotes exhaustive enumeration within these predefined structural families and circuit grid, not enumeration of every legal schedule DAG.

These fixed families serve two purposes. First, they provide explicit baselines throughout Chapter 6: the raw chain and the source-paper schedule provide reference points, while uniform link-level purification represents a simple non-adaptive baseline that does not require any search over schedule structure. Second, their candidates are always added to the candidates produced by

the two broader search tiers, so that these known schedule classes cannot be lost solely because of frontier pruning.

5.4 Span Dynamic Programming

For every span $\text{Span}(a, b)$, the second tier memoizes a Pareto frontier [69] of ways to create one resource over that span. A single-hop frontier ($b = a + 1$) contains the raw link and link-level purification choices. For a wider span, the recurrence tries every split point $a < m < b$ and combines each retained candidate from $\text{Span}(a, m)$ with each retained candidate from $\text{Span}(m, b)$ using a join operation. Each sub-span is computed once and reused wherever it appears in a wider construction. This reuse, rather than repeatedly constructing and evaluating complete end-to-end schedules, is the main computational advantage of the dynamic program.

The frontier retained at each span is multi-objective over $(C, F, P_{\text{success}})$ (Pareto dominance, Section 5.1). Only non-dominated candidates are retained, following the standard Pareto-pruning rule used in multi-criteria optimisation [71]. This pruning is exact with respect to the three quantities tracked by the frontier, but it does not imply that the resulting search is globally exhaustive once additional caps or heuristic retention rules are introduced.

5.5 Same-Span Entanglement Pumping

The span-partition recurrence alone constructs a resource by joining two different spans. It therefore does not represent repeated purification of a resource against a fresh resource at the same span. The second-tier extension adds this operation explicitly: given a candidate at span κ , it may consume that candidate together with an independently generated candidate at the same κ and retain the purified output at κ . The operation is bounded by the configured copy-count and circuit limits rather than allowing an unrestricted number of pumping rounds.

An uncapped mode is available for targeted validation. It restores exhaustive pumping within the configured circuit and copy-count limits, but can already become impractical for small N when the budget is large. It is therefore used as a diagnostic comparison rather than as the general production setting.

Pumping and ordinary join candidates compete for the same fixed-size frontier. Adding pumping can therefore remove a join-only candidate from a smaller span before that candidate can contribute to an end-to-end frontier. An option to disable pumping entirely reproduces the earlier pumping-free search behaviour. Section 6.3 reports both settings at the reference configuration because they can produce different top-ranked schedules.

5.6 Beam Search

The third tier, beam search, reuses the same span recursion, node construction, and evaluator as Section 5.4. It caps every retained span frontier at a beam width w_{beam} , with $w_{\text{beam}} = 25$ by default. This bounds runtime and memory at the network sizes studied in Chapter 6 (up to $N = 28$ hops), but a candidate that a wider beam would retain can be pruned before it reaches the end-to-end span. The cap is necessary in practice: an uncapped width-7 frontier already

contains roughly 1,200 candidates, making an uncapped search through the paper’s $N = 10$ chain impractical. The same w_{beam} also caps the same-span pumping pool, so pumping and ordinary join candidates share the available slots at each span.

At each span, the w_{beam} retained slots are divided between two rankings. Roughly half retain the highest-fidelity candidates, preserving purified building blocks that may only become useful after later joins. The remaining slots favour candidates that meet the fidelity hint F_{min} , then rank them by higher success probability, higher fidelity, and lower cost. This split matters because a purely efficiency-based ranking would systematically reject link-level purification at short spans, where a raw link is locally cheaper and more likely to succeed. Accumulated noise across several unpurified raw links can nevertheless later make the end-to-end fidelity floor unattainable, so retaining some higher-fidelity candidates guards against that failure mode.

Other heuristics could search the same space, but they would require a separate schedule encoding and proposal operators that preserve the **DAG** legality conditions of Section 4.3.1. The beam tier instead retains the span-partition representation and changes only how many candidates are kept at each frontier. This keeps all search tiers on the same physical evaluation path and makes repeated runs *deterministic*. The resulting schedule is nevertheless only a valid candidate found by a bounded search, not a guarantee of global optimality.

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